All these took place in what Sivan had famously described as "15 minutes of terror", the time taken for the lunar craft to break out of its orbit around the moon and land on the surface, a phase that is completely autonomous. Everything seemed to have been going fine with Chandrayaan-2 in what is known as the rough braking phase of the craft's descent. At the start of this descent, 30 km above the lunar surface, the spacecraft was going at 6,000 kmph, 3x the maximum speed of IAF's top-of-the-line jet fighter the Sukhoi Su30MKI. That speed had to be brought down to around 1,200 kmph (400 kmph more than a passenger jet's average speed of 800 kmph) and its height to around 8 km above the lunar surface. To do so, the spacecraft had to retrofire four of its engines to act as a braking mechanism and simultaneously command its cluster of eight thrusters to tilt the spacecraft at the angle needed for descent mode.

**On the consoles, the spacecraft** seemed to be doing exactly as it was meant to in this phase. In the next phase, known as the camera coasting or attitude hold phase, the spacecraft evaluates its position and decides if it needs course correction for the fine braking phase. Unknown to the mission team, things had started going wrong because the four thrusters over-delivered, reducing the spacecraft's speed by an additional 100 kmph. The navigation and guidance systems meant to compensate for the reduction in speed started an aggressive manoeuvre to steer it back on course. However, serious software errors began to occur at this point, confusing the spacecraft and even making it do an unplanned somersault. Meanwhile, instead of the engine thrust getting lowered, it was boosted to the maximum as the spacecraft's autonomous guidance and control systems desperately tried to correct the anomalies. By then, it was already too late, and the lander rammed into the moon's surface at a speed of 180 kmph, overshooting the designated spot by 500 metres. As a senior ISRO scientist puts it, "Instead of a normal 3-Sigma dispersion that the craft could have taken care of, we were dropping to sigma levels way beyond the capacity of the control systems to handle major deviations."

## THE FIGHTBACK

It is to the credit of the ISRO failure review system that it successfully pinpointed the root cause of the problems without allowing any acrimony to develop between the various teams. Clearly, every major system at the landing stage was accountable for the errors reaching unmanageable proportions, necessitating a complete overhaul of the spacecraft. In what was a saving grace for Team ISRO, Chandrayaan-2's orbiter was performing optimally, sending continuous data about the moon's surface and contributing immensely to Chandrayaan-3's success.



**WHILE INVESTIGATING CHANDRAYAAN-2'S FAILURE, PROBLEMS WERE FOUND WITH ITS PROPULSION, NAVIGATION AND GUIDANCE CONTROL**

Knowing what caused Chandrayaan-2's failure was only half the battle won. The tougher part was overcoming the mistakes. Veeramuthuvel, who was associate project director of Chandrayaan-2 and had gained immense understanding of the systemic issues involved, was made project director for Chandrayaan-3 in December that year. Kalahasti Kalpana, 49, who too had worked on Chandrayaan-2, was made his associate project director. Kalpana, who graduated from a Chennai engineering college, had joined ISRO as a radar engineer in 1999. Her experience reflects the ISRO culture of fostering gender equality, and Kalpana says, "I have never felt we are not equal, or discriminated against, because of my gender at ISRO. In all discussions and tasks, we are treated on par with men."